INCH-POUND

MIL-C-70957 (AR)
27 February 1992

MILITARY SPECIFICATION

CUSHION COMPONENTS FOR METAL CONTAINER FOR 105MM HOWITZER AMMUNITION

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions and packaging for one type of cushion components used for 105MM Howitzer ammunition (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-1752 - Cushioning Materials, Packaging, Polyethylene Foam

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

MIL-A-70625 - Automated Acceptance Inspection Equipment Design Testing and Approval of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8140

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

12934903	-	Container, PA117, and Cushion Assembly for 105MM Howitzer Cartridge, Packed in PA111
12934904	-	Container, PA117, and Cushion Assembly for 105MM Howitzer Cartridge, Packed in PA55 or M105A3
12934906	-	Support
12934907	-	Cushion, Bottom
12934908	-	Spacer
12934909	-	Cushion

(Copies of other Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the Solicitation (see 6.2).

AMERICAN SOCIETY FOR TEST AND MATERIALS

ASTM-D3951 - Commercial Packaging, Standard Practice for (DoD adopted)

(Applications for copies for ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all the requirements specified on drawings 12934903, 19234904, 12934906, 12934907, 12934908, and 12934909 and with associated drawings and specifications referenced thereon.

3.3 Compressive strength. The compressive strength and performance requirements of cushion and bottom cushion shall be as specified in PPP-C-1752 for Type V material.

3.4 First article inspection. When specified in the contract (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.5 Workmanship. The components shall be regular, smooth and free from any creases, wrinkles, cracks, rips, tears, dirt and any other defect which might offset the serviceability, durability, safety and appearance of the assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-A-48078 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of ten (10) each of every component.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

MIL-C-70957 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Components		See below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Support (Dwg. 12934906)			
	Examination for Defects	10	3.2	4.4.2.1
	Bottom Cushion, (Dwg. 12934907)			
	a. Examination for defects	10	3.2	4.4.2.2
	b. Compressive strength (see Note)	5	3.3	4.5
	Spacer (Dwg. 12934908)			
	Examination for defects	10	3.2	4.4.2.3
	Cushion (Dwg. 12934909)			
	a. Examination for defects	10	3.2	4.4.2.4
	b. Compressive strength (see Note)	5	3.3	4.5

NOTES:

The test for compressive strength shall be performed after examination for defects.

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials, the same material batch, and the same methods, in accordance with the same drawing, same drawing revision, same specification revision and manufactured under essentially the same conditions, and at essentially the same time. All materials submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. In addition, inspection lots of containers shall contain material from lots with the same interfix number from one manufacturer.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. When cited herein, attribute sampling inspection shall be in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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Table II. Attribute sampling inspection

Lot Size	Inspection Level I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether contractor's proposed alternative(s) provide equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70957 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Support	1 OF 1	12934906
			NEXT HIGHER ASSEMBLY: 12934903, 12934904

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification (see note)		3.2	Certification
102	Outside diameter	Level IV	3.2	Gage
103	Inside diameter	Level IV	3.2	Gage
104	Overall length, minimum (min)	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

Submit material certification with each lot submitted (see 6.3).

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70957 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Bottom cushion	1 OF 1	12934907
			NEXT HIGHER ASSEMBLY: 12934903, 12934904

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification (see Note)		3.2	Certification
102	Thickness	Level III	3.2	Gage
103	Compressive strength of material	5-0-1	3.3	Gage
104	Diameter	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

Submit material certification with each lot submitted (see 6.3).

AMSMC Form 1570b, 1 Jul 89 Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Spacer	1 OF 1	12934908
			NEXT HIGHER ASSEMBLY 12934904

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification (see Note)		3.2	Certification
102	Outside diameter	Level IV	3.2	Gage
103	Overall length, min.	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:
Submit material certification with each lot submitted (see 6.3).

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CLASSIFICATION OF CHARACTERISTICS

MIL-C-70957 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cushion	1 OF 1	12934909
			NEXT HIGHER ASSEMBLY: 12934903, 12934904

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material certification (see Note)		3.2	Certification
102	Thickness	Level III	3.2	Gage
103	Compressive strength of material	5-0-1	3.3	Gage
104	Diameter	Level IV	3.2	Gage
105	Color incorrect or absent	Level IV	3.1	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.5	Visual

NOTES:

Submit material certification with each lot submitted (see 6.3).

AMSMC Form 1570b, 1 Jul 89

Replaces 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Container, PA117, and Cushion Assembly for Cartridge, 105MM Howitzer Cartridge, Packed in PA111 (prior to closing)	1 OF 1	12934903
			NEXT HIGHER ASSEMBLY 12934900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Adhesive on gasket surface or mating body surface of container	100%	3.2	Visual
102	Packing components incorrectly assembled	Level III	3.2	Visual
103	Packing components missing or damaged	Level III	3.2	Visual
Minor				
201	Protective finish missing, damaged or bare spots exposing metal	Level IV	3.5	Visual
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Container, PA117, and Cushion Assembly for Cartridge, 105MM Howitzer Cartridge, Packed in PA55 or M105A3 (prior to closing)	1 OF 1	12934904
			NEXT HIGHER ASSEMBLY 12934901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Adhesive on gasket surface or mating body surface or container	100%	3.2	Visual
102	Spacer missing	100%	3.2	Visual
103	Packing components incorrectly assembled	Level III	3.2	Visual
104	Packing components missing or damaged	Level III	3.2	Visual
Minor				
201	Protective finish missing, damage, or bare spots exposing metal	Level IV	3.5	Visual
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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4.4.3 Testing.

4.4.3.1 Compressive strength (see 3.3) Major defect. Three (3) samples per material lot shall be selected prior to start of production. If any sample fails to meet the requirements, the lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with 4.4.2.1 and in 4.5. When Automated Acceptance Inspection Equipment is to be used, the provisions of MIL-A-70625 shall apply. The contractor shall submit, for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection. (see 6.5)

4.5.1 Compressive strength. The compressive strength of the cushion shall be determined in accordance with PPP-C-1752.

5. PACKAGING

5.1 Preservation and packaging. None required.

5.2 Packing.

5.2.1 Commercial. The cushion components and container assemblies shall be packed in accordance with ASTM-D3951.

5.3 Marking.

5.3.1 Commercial. The containers and cushion components shall be marked in accordance with ASTM-D3951 and the following additional instructions. Commercial shipping tags, attached with wire, shall be used for metal containers unless otherwise specified. The following shall be shown: Item name and model number, part number and revision, contract number, lot number and date, name and address of manufacturer. If components and assemblies are bundled, boxed or palletized, the gross weight and quantity shall be included. The above information may be placed on pallets or boxes by labeling or stenciling.

5.4 Shipping. When containers, cushion assemblies and cushion components from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

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6. NOTES

(This section contains information of general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. These containers and cushion components are intended for use as the packaging and cushioning for 105MM Howitzer Cartridges.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Applicable stock number.
d. First article sample requirements.
e. Packaging requirements, if other than specified in Section 5.
f. Serialization requirements, if applicable.
g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.4 Submission of contractor inspection equipment for design approvals. See MIL-A-48078. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be as specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

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6.6 Subject term (keyword listing).

Cartridges
Packing
Pallet

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 8140-A810)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-70957 (AR)	27 FEBRUARY 1992

3. DOCUMENT TITLE
CUSHION COMPONENTS FOR METAL CONTAINER FOR 105MM HOWITZER AMMUNITION

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (*Include Zip Code*) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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